

PROFESSIONAL EXPERIENCE

2022–Now ML Scientist

Yoshua Bengio's Group, Mila Québec AI Institute [↗](#), QC/CA.

- Lead developer of [TorchGfN](#) [↗](#), the reference open-source GFlowNet library (312 stars, 56 forks, 20 contributors). Designed the library's architecture, contributed substantial code, review all contributions, maintain the library.
- Industrial research collaborations: antibody design with [Amgen](#) [↗](#), large-scale distributed training of GFlowNets with [Intel](#) [↗](#).
- Initiated and wrote the grant proposal for a multi-centre AI for Metabolic Engineering collaboration with the US Department of Energy's [Joint BioEnergy Institute](#) [↗](#); co-authored the proposal for Mila's national compute infrastructure.
- ML engineering support for the lab's AI for Science codebases, delivering the training efficiency required for projects such as [RGfN](#) [↗](#) (drug discovery) and [FLECS](#) [↗](#).

2021–2022 ML Research Scientist

Deep Genomics [↗](#), QC/CA.

- Architectures for sequence-based RNA foundation models, and multi-modal methods allowing sequence-to-sequence models to generalize to unseen cell types — a core bottleneck in RNA therapeutic discovery.
- Built the anomaly detection pipeline for the drug screening platform, enabling early detection of laboratory failures.

2021–2025 Scientific Mentor (Volunteer)

Creative Destruction Lab Montréal [↗](#), QC/CA.

- Technical due diligence and mentorship for early-stage deep-tech startups in the NLP, computer vision, and ML for biology streams, presented to a founder and investor audience.

2020–2021 Applied Research Scientist

Mila Technology Transfer, Mila Québec AI Institute, QC/CA.

- Scoped, proposed, and delivered applied deep learning projects for Mila's industrial partners: an equity momentum trading strategy with [CDPQ](#) [↗](#) (C\$450B pension fund) and digital forgery detection with [Jumio](#) [↗](#).

2020 PhD to VC Fellow

Fifty Years [↗](#), CA/US.

- Selective fellowship for technical PhDs: sourced pre-seed and seed deep-tech companies, performed technical due diligence, and helped source technical talent.

2020 Research Intern

Google, PA/US.

- Epistemic uncertainty estimation for the search ads predicted click through rate team.

2019 Research Intern

Imagia, QC/CA.

- Cancer localization without segmentation labels, using clinical notes as weak supervision — addressing the label scarcity that blocks clinical deployment of medical imaging models.

2014–2017 Research Methods Specialist

Centre for Addiction and Mental Health, ON/CA.

- Imaging biomarkers for [schizophrenia](#) [↗](#) and [Alzheimer's disease](#) [↗](#), contributing to 2 funded grants and 15 publications.
- Designed and ran (team of 5) a [data management platform](#) [↗](#), a 22-node compute cluster, and [QA tooling](#) [↗](#) used daily by 20 researchers across a multi-site consortium.

OPEN SOURCE

2022–Now TorchGfN [↗](#)

Lead developer & maintainer.

- Modular GFlowNet library: 312 stars, 56 forks, 20 contributors. The reference implementation for GFlowNet research, used researchers in academia and industry. [JMLR, 2026](#) [↗](#).

2020–2022 [TorchXRayVision](#) [↗](#)

Core contributor.

- Library of chest X-ray datasets and pre-trained models: 1,180 stars, 255 forks. [PMLR, 2022](#) [↗](#).

AWARDS

2026 **Canada Graduate Scholarships – Doctoral**

NSERC.

Canada's most competitive graduate scholarship. \$40,000/year for 3 years.

2026 **Gold Reviewer Award**

ICML [↗](#) Position Paper Track.

Awarded to the top-rated reviewers of the track.

EDUCATION

2026– **PhD, Computer Science (incoming)**

[Mila](#) [↗](#), Université de Montréal, QC/CA.

- Supervised by [Dr. Pierre Luc Bacon](#) [↗](#)
Topic: Sample Efficient RL for High Dimensional Search.
Funded by the NSERC Canada Graduate Scholarship – Doctoral.

2018–2020 **MSc., Computer Science**

[Mila](#) [↗](#), Université de Montréal, QC/CA.

- Supervised by [Dr. Yoshua Bengio](#) [↗](#)
'Methods for controlling and utilizing saliency maps in medical imaging' [↗](#).

2011–2013 **MSc. with Distinction, Biology**

York University, ON/CA.

- Thesis Supervised by [Dr. Keith Schneider](#) [↗](#)
'Tremotopic mapping of the human thalamic reticular nucleus'.

2005–2009 **BSc. Hons., Psychology**

Queen's University, ON/CA.

INVITED TALKS

2024 **Keynote Speaker**

FACSS SciX.

Can an AI Understand a Scientist?

2024 **Invited Speaker**

Faculty of Pharmacy, University of Montreal.

Machine Learning for Drug Discovery and Development.

2023 **Invited Speaker**

Joint Genome Institute Annual User Meeting.

Considerations for Machine Learning-Aided Lab in the Loop Scientific Discovery.

COMMUNITY SERVICE

2023–Now **Area Chair**

Meta Reviewing.

Paper adjudication for the [NeurIPS Datasets and Benchmarks Track](#) [↗](#).

2014–Now **Reviewing**

Paper reviews for conferences and journals.

Reviews for [ICLR](#) [↗](#), [TMLR](#) [↗](#), [AAAI](#) [↗](#), [NeurIPS](#) [↗](#), [ICML](#) [↗](#), [MIDL](#) [↗](#), [PNAS](#) [↗](#), [Biological Psychiatry](#) [↗](#), & [PLOS ONE](#) [↗](#).

2023 **Grant Reviewing**

Health Data Nexus Dataset Grants.

Adjudicated 5 proposals on the collection of healthcare-focused ML-ready datasets.

2026 **Instruction**

Introduction to Agentic Engineering.

A 3-hour workshop for undergraduate and Master's level students, [Techaide](#) [↗](#).

2019 **Instruction**

Deep Learning for Neuroscientists.

Deep learning fundamentals for a life science audience, at [MAIN 2019](#) [↗](#) and [Brainhack 2019](#) [↗](#).

SELECTED PUBLICATIONS

*h-index 20, 2375 citations. * denotes equal contribution. Also on [Google Scholar](#).*

Richardson OE, Samiei M, Shakerinava M, **Viviano JD**, El Kabid A, Parviz A, Bengio Y. Local Inconsistency Resolution: The Interplay between Attention and Control in Probabilistic Models. International Conference on Artificial Intelligence and Statistics (AISTATS), Spotlight, 2026.

Viviano JD*, Younis OG*, Choi S*, Schmidt V, Bengio Y, Lahlou S. torchgfn: A PyTorch GFlowNet library. Journal of Machine Learning Research (JMLR), In Press, 2026.

Boussif O, Ezzine LN, **Viviano JD**, Koziarski M, Jain M, Malkin N, Bengio E, Assouel R, Bengio Y. Action abstractions for amortized sampling. International Conference on Learning Representations (ICLR), 2025.

Bertin P, **Viviano JD**, Tejada-Lapuerta A, Wang W, Bauer S, Theis FJ, Bengio Y. A scalable gene network model of regulatory dynamics in single cells. arXiv, 2025.

Delaunay P, Bouthillier X, Breuleux O, Ortiz-Gagné S, Bilaniuk O, Normandin F, Bergeron A, Carrez B, Alain G, Blanc S, Osterrath F, **Viviano JD**, et al. Introducing Milabench: Benchmarking Accelerators for AI. arXiv, 2024.

Cohen JP, **Viviano JD**, Bertin P, Morrison P, Torabian P, Guarrera M, Lungren MP, Chaudhari A, Brooks R, Hashir M, Bertrand H. TorchXRyVision: A library of chest X-ray datasets and models. International Conference on Medical Imaging with Deep Learning (MIDL), 2022.

Oliver LD, Hawco C, **Viviano JD**, Voineskos AN. From the Group to the Individual in Schizophrenia Spectrum Disorders: Biomarkers of Social Cognitive Impairments and Therapeutic Translation. Biological Psychiatry, 91(8), 699-708, 2022.

Cohen JP, Cao T, **Viviano JD**, Huang CW, Fralick M, Ghassemi M, Mamdani M, Greiner R, Bengio Y. Problems in the deployment of machine-learned models in health care. Canadian Medical Association Journal (CMAJ), 193(35), E1391-E1394, 2021.

Luccioni A, **Viviano JD**. What's in the Box? A Preliminary Analysis of Undesirable Content in the Common Crawl Corpus. The Joint Conference of the Association for Computational Linguistics and the International Joint Conference on Natural Language Processing (ACL-IJCNLP), 2021.

Viviano JD, Simpson B, Dutil F, Bengio Y, Cohen JP. Saliency is a Possible Red Herring When Diagnosing Poor Generalization. International Conference on Learning Representations (ICLR), 2021.

Overton DJ, Bhagwat N, **Viviano JD**, Jacobs GR, Voineskos AN. Identifying psychosis spectrum youth using support vector machines and cerebral blood perfusion as measured by arterial spin labeled fMRI. Neurolmage: Clinical, 27, 102304, 2020.

Hawco C, Buchanan RW, Calarco N, Mulsant BH, **Viviano JD**, Dickie EW, Argyelan M, Gold JM, Iacoboni M, DeRosse P, Foussias G, Malhotra AK, Voineskos AN, for the SPINS group. Separable and replicable neural strategies during social brain function in people with and without severe mental illness. American Journal of Psychiatry, 176(7), 521-530, 2019.

Viviano JD, Buchanan RW, Calarco N, Gold J, Foussias G, Bhagwat N, Stefanik L, Hawco C, Malhotra AK, Voineskos AN, for the SPINS group. Resting-state connectivity biomarkers of cognitive performance and social function in schizophrenia spectrum disorders and healthy controls. Biological Psychiatry, 84(9), 665-674, 2018.

Bhagwat N, **Viviano JD**, Voineskos AN, Chakravarty MM. Modeling and prediction of clinical symptom trajectories in Alzheimer's disease using longitudinal data. PLOS Computational Biology, 14(9), e1006376, 2018.

Spreng RN, Stevens WD, **Viviano JD**, Schacter D. Attenuated anticorrelation between the default and dorsal attention networks with aging: Evidence from task and rest. Neurobiology of Aging, 45, 149-160, 2016.

DeSimone K, **Viviano JD**, Schneider KA. Population receptive field estimation reveals new retinotopic maps in human subcortex. Journal of Neuroscience, 35(27), 9836-47, 2015.

Viviano JD, Schneider KA. Interhemispheric interactions of the human thalamic reticular nucleus. Journal of Neuroscience, 35(5), 2026-32, 2015.